



HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

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Quiz # 3 Solution

QUESTION 1: HASHING (15 marks)

- a. The Balkan countries are some of the most underrated European nations. They have the same beauty as that of their neighboring countries, yet, people rarely travel over there. In order to boost their tourism, they decide to do some analysis, by gathering which visitor travelled to which Balkan country the most. Each country provides a list of tourists in the past 5 years. If a person has visited more than once, his/her name will appear in the list multiple times. As there are 11 Balkan states, you will have 11 lists, but for your convenience, they have all been merged into 1 single list, which means you will only be accessing one list. You are required to calculate the number of times a person has visited the Balkan states. Write an algorithm using Hashing that populates the hash table in such a way that the name represents the “key” and his/her number of trips to all the Balkan countries, represents the “value”. The size of hash table is taken as input from the user. Hash function to be used is the sum of ASCII values of the letters in the person’s name (exactly like you have done in the labs), and collisions must be handled using Quadratic Probing. You may use two separate lists to represent keys and values just like you did in the labs. **YOU ARE NOT ALLOWED TO USE ANY ADDITIONAL DATA STRUCTURES, APART FROM THE ONES SPECIFIED IN THE QUESTION.**

Solution:

- Size = take input from the user
- Keys = [None] * Size
- Values = [None] * Size
- name_lst → given in the question
- for each element, i in name_lst, do the following steps
 - sum1 = 0
 - for letter in i, i.e., name from the name_lst, calculate its ASCII sum as follows:
 - sum1 += ord(i)
 - slot_index = sum1 mod Size
 - if Keys[slot_index] == None, do as follows:
 - Keys[slot_index] = i
 - Values[slot_index] = 1
 - Else if Keys[slot_index] == i, i.e., the element is already added, then we only need to increase the frequency count in values as follows:
 - Values[slot_index] += 1
 - Otherwise, there might be collision because of slot being already occupied by some other number, so handle collisions as follows:
 - iteration = 1
 - new_index = slot_index
 - while (Keys[new_index] != None and iteration <= Size), resolve collision as follows:
 - ❖ new_index = (slot_index + (iteration**2)) mod Size
 - ❖ if Keys[new_index] == None, then we have found a new empty slot so update as follows:
 - Keys[new_index] = i
 - Values[new_index] = 1
 - Break
 - ❖ Else if Keys[new_index] == i, i.e., the same element matches then increase its frequency in Values list as follows:
 - Values[new_index] += 1
 - Break
 - ❖ Otherwise, move to the next slot to find empty spot
 - iteration = iteration + 1

b. Using the algorithm from part (a), perform hashing on the following list of names from all the 11 countries.

Assume size of hash table is 8.

[‘Sam’, ‘Mars’, ‘Eddie’, ‘Eddie’, ‘Eddie’, ‘Bob’, ‘Sam’, ‘Mars’, ‘Zack’, ‘Sam’, ‘Sam’, ‘Zack’, ‘Zack’, ‘Mars’, ‘Leo’, ‘Nick’, ‘Kate’, ‘Bob’, ‘Zack’, ‘Eddie’, ‘Nick’, ‘Sam’, ‘Leo’, ‘Mars’, ‘Kate’, ‘Kate’, ‘Kate’]

For your convenience, the sum of ASCII characters for the given names are already provided. You may use them directly to calculate the slot index:

- ‘Sam’ = 289
- ‘Mars’ = 403
- ‘Eddie’ = 475
- ‘Bob’ = 275
- ‘Zack’ = 393
- ‘Leo’ = 288
- ‘Nick’ = 389
- ‘Kate’ = 389

Solution: Initially hashtable is empty and filled with None

<u>Keys:</u>	None	None	None	None	None	None	None	None
<u>Values:</u>	None	None	None	None	None	None	None	None

- Sam: $289 \bmod 8 = 1$
- Mars: $403 \bmod 8 = 3$
- Eddie: $475 \bmod 8 = 3 \rightarrow$ collision, so do quadratic probing:
 - $(3 + 1^2) \bmod 8 = 4 \bmod 8 = 4$, which is free
- Bob: $275 \bmod 8 = 3 \rightarrow$ collision, so apply quadratic probing until free slot is found:
 - $(3 + 1^2) \bmod 8 = 4 \bmod 8 = 4 \rightarrow$ collision again, so continue to probe
 - $(3 + 2^2) \bmod 8 = 7 \bmod 8 = 7$, which is free
- Zack: $393 \bmod 8 = 1 \rightarrow$ collision, so do quadratic probing:
 - $(1 + 1^2) \bmod 8 = 2 \bmod 8 = 2$, which is free
- Nick: $389 \bmod 8 = 5$
- Leo: $288 \bmod 8 = 0$
- Kate: $389 \bmod 8 = 5 \rightarrow$ collision, so do quadratic probing:
 - $(5 + 1^2) \bmod 8 = 6 \bmod 8 = 6$, which is free

The resultant hashtable:

<u>Keys:</u>	Leo	Sam	Zack	Mars	Eddie	Nick	Kate	Bob
<u>Values:</u>	2	5	4	4	4	2	4	2

QUESTION 2: GRAPH TRAVERSAL (10 marks)

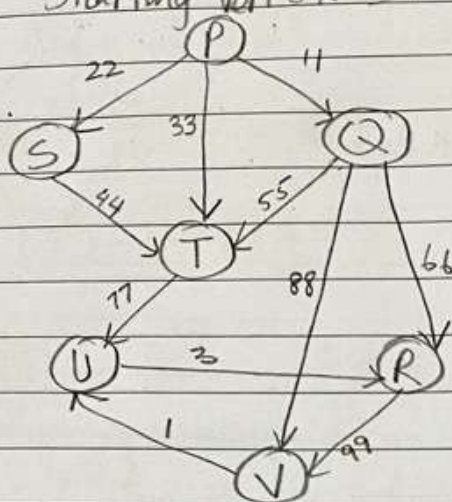
For the given graph, apply the Breadth-First Search Traversal Algorithm with the starting vertex = P. Show all the working for each step. Construct the corresponding BFS graph, by showing all the tree edges and non-tree edges. Use different notation to represent each edge, and specify a legend for it. Follow ascending order in choosing vertices. Calculate the total cost of traversal in the end.

IF YOU ARE MISSING ANY INTERIM CALCULATIONS, YOU WILL NOT GET ANY MARKS

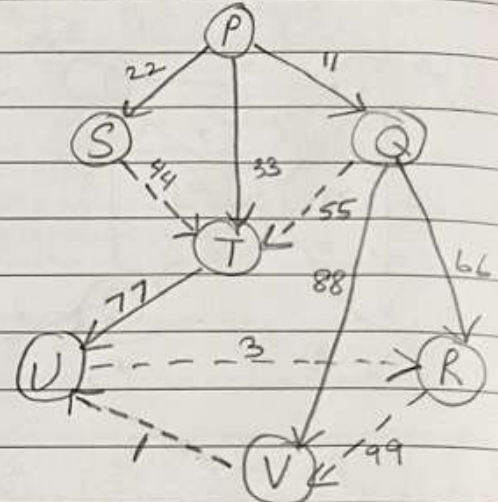
Solution:

Date:

BFS: Starting vertex = P



Legend:
 → Tree Edge
 --- Back Edge
 --- Forward Edge



Queue

Visited List

Path

[P, S, T]

[P*]

[S, T, R, T, V]

[P*, Q]

P → Q? ✓

[T, R, T, V, T]

[P*, Q, S]

P → S? ✓

[R, T, V, T, U]

[P*, Q, S, T]

P → T? ✓

[T, V, T, U, V]

[P, Q*, S, T, R]

Q → R? ✓

[X, T, U, V]

"

Q → T? X

[T, U, V, U]

[P, Q*, S, T, R, V]

Q → V? ✓

[X, V, U]

[P, Q, S*, T, R, V]

S → T? X

[X, U, R]

[P, Q, S, T*, R, V, U]

T → U? ✓

[X, R]

[P, Q, S, T, R*, V, U]

R → V? X

[R]

[P, Q, S, T, R, V*, U]

V → U? X

[]

[P, Q, S, T, R, V, U*]

U → R? X

Path: P → Q, P → S, P → T, Q → R, Q → V, T → U
Cost: 11 + 22 + 33 + 66 + 88 + 77
 = 297

QUESTION 3: SHORTEST PATH (15 marks)

- a. Run the Dijkstra's Algorithm on the following undirected, weighted graph, to get the shortest path of every vertex from the starting vertex, D. Specify shortest path of every vertex from the starting vertex, and calculate the cost of this shortest path. Draw the reduced graph after Dijkstra's algorithm has successfully been implemented, showing the shortest paths between all the vertices. Show your complete working. **IF YOU DO NOT SHOW ANY WORKING, YOU'LL NOT GET MARKS FOR IT**

Solution:

Date: ,

Dijkstra: Starting vertex = D

Visited	A	B	C	D	E	F	G
D	∞	∞	∞	0	∞	∞	∞
B	∞	2	∞		2	3	∞
E	6		3		2	3	∞
C	6		3			3	9
F	6					3	9
A	6						8
G							8

Src $\rightarrow$ Dest	Shortest Path	Cost
D $\rightarrow$ A	A, B, D $\Rightarrow$ D $\rightarrow$ B $\rightarrow$ A	2 + 4 = 6
D $\rightarrow$ B	B, D $\Rightarrow$ D $\rightarrow$ B	2
D $\rightarrow$ C	C, B, D $\Rightarrow$ D $\rightarrow$ B $\rightarrow$ C	2 + 1 = 3
D $\rightarrow$ E	E, D $\Rightarrow$ D $\rightarrow$ E	2
D $\rightarrow$ F	F, D $\Rightarrow$ D $\rightarrow$ F	3
D $\rightarrow$ G	G, F, D $\Rightarrow$ D $\rightarrow$ F $\rightarrow$ G	3 + 5 = 8

Reduced graph on top besides the Question

DALMATON

b. For the same graph, as given in part (a), define the Graph using Adjacency List Representation

Solution:

$G(V, E) = \{$ A : [(A, B, 4), (A, C, 3)],
B : [(B, A, 4), (B, C, 1), (B, D, 2)],
C : [(C, A, 3), (C, B, 1), (C, E, 4)],
D : [(D, B, 2), (D, E, 2), (D, F, 3)],
E : [(E, C, 4), (E, D, 2), (E, F, 2), (E, G, 7)],
F : [(F, D, 3), (F, E, 2), (F, G, 5)],
G : [(G, E, 7), (G, F, 5)]
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